

Lecture 02

Evaluation Metrics – part 1

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Today's Class

- Evaluation Metrics for Recommender Systems
 - Evaluation paradigms: user studies, online evaluation and offline evaluation
 - Design issues in offline recommender evaluation
 - Accuracy metrics in offline evaluation
 - Limitations of evaluation measures



Why Do We Learn Evaluation Metrics?

- **Enhancing User Experience**

- One of the primary goals of recommender systems is to improve user satisfaction
- Through evaluation, we can determine which recommender system provides valuable items to actual users, allowing us to identify ways to improve the system from the user's perspective

- **Model Selection**

- Recommender systems can be implemented using various algorithms and approaches
- To determine which algorithm is most effective, it is necessary to have methods for evaluating and comparing different algorithms

WHICH RECOMMENDER SYSTEM IS BETTER?

Why Do We Learn Evaluation Metrics?

- **Multiple Facets**

- Evaluating recommender systems is often multifaceted, and a single criterion cannot capture the diverse goals of the designer

- **Appropriate Design for Evaluation**

- A well-designed evaluation system is crucial for understanding the effectiveness of various recommendation algorithms
- An improperly designed experimental evaluation can lead to either a significant underestimation or overestimation of the true accuracy of a particular algorithm or model

Two Categories of Evaluation Methods

- **Online Methods**

- User reactions are measured with respect to the presented recommendations
- User participation is essential in online systems
- However, since online evaluations require active user participation, **they are often not feasible for benchmarking and research**

Two Categories of Evaluation Methods

- **Offline Methods**

- **Historical data sets** are used for offline evaluations
- Offline methods are by far the most common approach for evaluating recommender systems from both research and practical perspectives

Issues with Offline Methods

- **Accuracy Metrics**

- Accuracy metrics remain the most important component in the evaluation

- **Experimental Design**

- It is crucial to design experiments carefully to avoid overestimating or underestimating accuracy
- Careful experimental design is essential

- **Evaluation Goals**

- In addition to accuracy metrics, secondary goals such as novelty, trust, coverage, and serendipity are important for the user experience



Evaluation Paradigms

- There are three primary types of evaluation for recommender systems: **user studies**, **online evaluations**, and **offline evaluations** using historical data sets

Evaluation Paradigms

- User Studies

Questions	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree	Acceptable answer
		Serial number	Questions			
The Recommendations are relevant to my Activities	☐	1	What is your opinion about online doctor-recommender system idea?			Useful/Not useful
The Recommendations are Surprising to me	☐	2	Have you ever used any online healthcare plate form? If yes can you tell any deficiencies in those systems?			User answer/Free answer
The Recommendations differ significantly from each other	☐	3	What kind of doctor's attributes you look for while seeking an appropriate doctor? Enlist them			User answer/Free answer
The Recommendations are useful to me	☐	4	Enlist sources from which you generally get appropriate doctor's recommendations			User answer/Free answer
I am satisfied with the language of recommendations	☐	5	In what ways doctor recommendation will			User answer/Free
The recommendations provide me with novel information	☐					

Evaluation Paradigms

- **User Studies**

- Test subjects are actively recruited and asked to interact with the recommender systems to perform specific tasks
- Feedback can be collected from users both before and after their interaction, while the system also gathers information about their interactions with the recommender systems

- **Advantages**

- Information about user interactions with the system can be collected
- Various scenarios can be tested to assess the impact of changes to the recommender system on user interactions, such as the effect of changing a particular algorithm or user interface

Evaluation Paradigms

- User Studies

- Disadvantages

- Users' active awareness of the recommender system testing can often **bias** their choices and actions

성격 유형 퀴즈

친구들이 보는 내 성격



Evaluation Paradigms

- User Studies

- Disadvantages

- Users' active awareness of the recommender system testing can often **bias** their choices and actions
 - Recruiting large cohorts of users for evaluation purposes is also **difficult** and **expensive**

[원내 게시용]

건강기능식품 연구 참여자 모집

신선초추출물의
간기능 개선효과 및
안전성 평가를 위한 인체적용연구

연구명칭 및 목적:
알코올 섭취로 간기능이 저하된 성인을 대상으로 신선초추출물의 간기능 개선효과와 안전성 평가
* 신선초는 영일엽이라고도 불리며 오랜 식경험이 있는 식물이며, 식용원료로 사용이 가능합니다.

참여대상
1) 만 20세 이상 75세 미만의 성인 남녀
2) GGT가 정상 상한치를 초과하는 분
3) ALT, AST 중 한가지 이상이 정상 상한치의 1.5배 이상인 분
4) 최근 1년 이상 다음과 같이 알코올을 섭취해온 분
— 남성: 210 g/주(소주로 약 4병/주) 이상
— 여성: 105 g/주(소주로 약 2병/주) 이상
* 단, 알코올 관련 정신/행동장애로 진단받았거나 또는 알코올이 아닌 다른 이유로 발생한 간질환 등이 있는 분들은 본 연구에 참여하실 수 없습니다.

모집 인원: 88명

연구방법
— 선별검사 후, 시험군 또는 대조군에 무작위 배정되어 12주간 시험식품 섭취
— 검사항목: 혈액검사, 신체검사, 설문조사 등
— 방문횟수(연구참여기간 총 3회(약 12주))

참여 시 제공사항:
— 소정의 교통비
— 연구참여기준에 적합하신 분에 한해서 혈액소변검사 결과 제공

예측 가능한 부작용
연구에 사용되는 '신선초추출물'에 대해 특별히 알려진 부작용은 없으나, 연구참여로 인해 예상치 못하게 발생하는 불편감 또는 이상반응에 대해 신속하고 적절한 조치로 귀하의 안전에 만전을 기할 것입니다.

신청 및 문의
— 연구책임자: 차의과학대학교 분당차병원 간내과 황성규 교수

☎ 신청 및 문의 031-780-6261
※ 참여를 위한 자세한 사항은 위 연락처로 연락 부탁드립니다.

차의과학대학교 분당차병원 경기도 성남시 분당구 아람로 500-150 | 연구자: (주)비인조바이오텍(경기 성남시 조계로 30, 02-400-2009)

Evaluation Paradigms

- **User Studies**

- **Disadvantages**

- Users' active awareness of the recommender system testing can often **bias** their choices and actions
 - Recruiting large cohorts of users for evaluation purposes is also **difficult** and **expensive**
 - Recruited users are not always representative of the general population because the recruitment process itself introduces biases that cannot be fully controlled
 - The results from user evaluations may not be entirely reliable

Evaluation Paradigms

- **Online Evaluation**

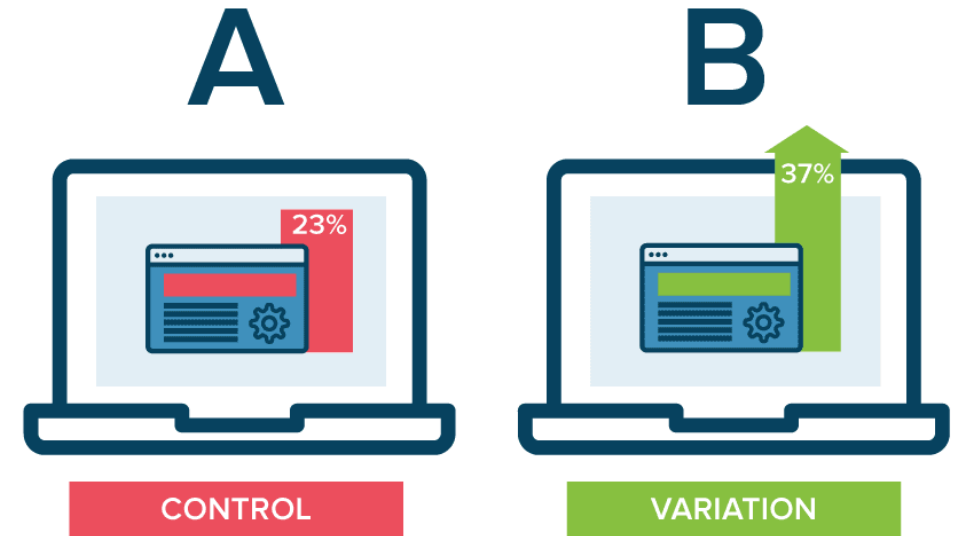
- Online evaluations also involve user studies, but in this case, the users are typically real users interacting with a fully deployed or commercial system
- This approach is sometimes less susceptible to recruitment bias because the users are directly interacting with the system as part of their natural activities
- Users can be sampled randomly, and various algorithms can be tested with each user sample

Evaluation Paradigms

- **Online Evaluation**

- **A/B Testing**

- A/B testing measures the direct impact of the recommender system on end users
 - Divide the users into two groups, A and B
 - Apply one algorithm to group A and a different algorithm to group B for a set period, ensuring that all other conditions between the two groups are as similar as possible
 - At the end of the process, compare the target metrics (e.g., user retention, revenue, etc.) between the two groups



Evaluation Paradigms

- **Online Evaluation**

- **Advantages of A/B Testing**

- This method is the **most accurate** for testing the long-term performance of the system in terms of goals such as profit

- **Disadvantages of A/B Testing**

- Such systems cannot be realistically deployed unless a large number of users are already enrolled, making this method difficult to use during the startup phase
 - Such systems are typically **not openly accessible** and are **only available to the owner of the specific commercial system**
 - The tests are often **not generalizable** for system-independent benchmarking by researchers and practitioners

Evaluation Paradigms

- **Offline Evaluation**

- Offline testing utilizes historical data, such as ratings

- **Advantages**

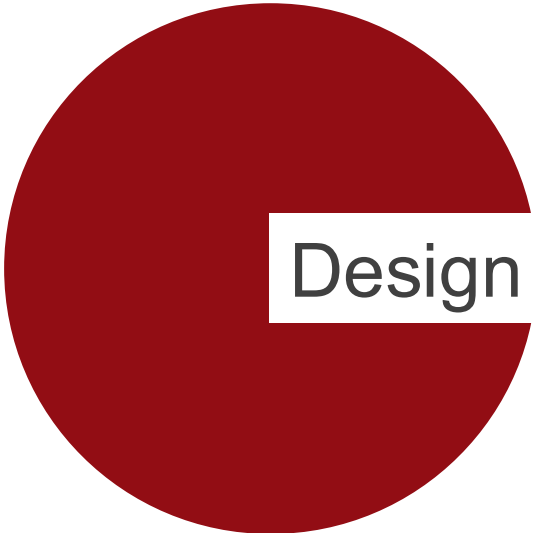
- One advantage of using historical data sets is that they **do not require access to a large user base**
- Once a data set is collected, it can serve as a standardized benchmark for comparing various algorithms across different settings
- Multiple data sets from various domains can be used to test the generalizability of recommender systems
- Standardized frameworks and evaluation measures have been developed for many scenarios

Evaluation Paradigms

- **Offline Evaluation**

- **Disadvantages**

- They do not assess the user's future propensity to react to the recommender system
 - Data may evolve over time, and current predictions might not reflect the most appropriate predictions for the future
 - Measures like accuracy do not capture important characteristics of recommendations, such as *serendipity* and *novelty*, which can significantly impact the long-term conversion rate of the recommendations



Design Issues in Offline Recommender Evaluation

Design Issues in Offline Recommender Evaluation

- Recap: Two Types of User Feedback

	5	4		2	2	
		2	3	5		4
	5	?	5	3	2	?

<Explicit Feedback>

	✓				✓	
		✓				✓
	?	✓	?	✓	?	?

<Implicit Feedback>

Design Issues in Offline Recommender Evaluation

- Relation with Machine Learning Problems

	5	4		2	2	
		2	3	5		4
	5	?	5	3	2	?

→ Regression!

<Explicit Feedback>

	✓				✓	
		✓				✓
	?	✓	?	✓	?	?

→ Classification!

<Implicit Feedback>

Design Issues in Offline Recommender Evaluation

- **Relation with Machine Learning Problems**

- Collaborative filtering can be viewed as a generalization of **classification** and **regression** modeling problem
- Therefore, many evaluation processes in recommender systems employ evaluation metrics from the field of machine learning
- Nevertheless, there are several aspects of the evaluation process that are **unique to collaborative filter applications**

Design Issues in Offline Recommender Evaluation

- **Preliminaries**

- Let $U = \{u_1, u_2, \dots, u_n\}$ and $V = \{v_1, v_2, \dots, v_m\}$ represent the set of users and items, respectively
- User-item interactions can then be represented as a matrix of size $n \times m$

	v_1	v_2	v_3	...	v_m
u_1					
u_2					
u_3					
u_4					
u_5					
...					
u_n					

<User-Item Interaction Matrix>

Design Issues in Offline Recommender Evaluation

• Preliminaries

- We refer to each user-item interaction as an entry
- For instance, an entry for user u_2 and item v_3 is denoted as $d = (u_2, v_3, 5)$, where 5 represents the rating given by user u_2 for item v_3
- In general, we denote entries as $d = (u_i, v_j, r_{i,j})$, where $r_{i,j}$ represents either the rating (e.g., 1, 2, 3, 4, or 5 in a 5-star rating system) or *is_click* indicator (e.g., *True* or *False* in implicit feedback)
- A dataset $D = \{d_1, d_2, d_3, \dots\}$ is a set of entries from U and V

	v_1	v_2	v_3	...	v_m	
u_1	2			3	3	
u_2		5	5			5
u_3		4	3			
u_4	5				4	
u_5		2		3		4
...	4		5			
			1	1		
		2			3	
u_n	2			4		5

<User-Item Interaction Matrix>

Design Issues in Offline Recommender Evaluation

- **Data Splits**

- In general machine learning problems, the dataset is randomly shuffled and divided into training and test sets



	v_1	v_2	v_3	$\dots$	v_m	
u_1	2			3	3	
u_2		5	5			5
u_3		4	3			
u_4	5				4	
u_5		2		3		4
$\dots$	4		5			
			1	1		
		2			3	
u_n	2			4		5

<User-Item Interaction Matrix>

Design Issues in Offline Recommender Evaluation

- **Data Splits**

- When the training set and test set are overlap, model performance may be overestimated

	v_1	v_2	v_3	...	v_m
u_1	2			3	3
u_2		5	5		5
u_3		4	3		
u_4	5				4
u_5		2		3	4
...	4		5		
			1	1	
		2			3
u_n	2			4	5

<User-Item Interaction Matrix>

Design Issues in Offline Recommender Evaluation

- **Data Splits**

- When the training set and test set are divided by rows or columns, model performance may be underestimated
- We refer to this issue as the cold-start problem

	v_1	v_2	v_3	...	v_m
u_1	2			3	3
u_2		5	5		5
u_3		4	3		
u_4	5				4
u_5		2		3	4
...	4		5		
			1	1	
		2			3
u_n	2			4	5

<User-Item Interaction Matrix>

Design Issues in Offline Recommender Evaluation

- **Data Splits**

- The dataset should be properly split to avoid overestimating or underestimating the model
- In other words, the test set should be designed to closely resemble the actual service environment

	v_1	v_2	v_3	...	v_m
u_1	2			3	3
u_2		5	5		5
u_3		4	3		
u_4	5				4
u_5		2		3	4
...	4		5		
			1	1	
		2			3
u_n	2			4	5

<User-Item Interaction Matrix>

Design Issues in Offline Recommender Evaluation

'직계궤 외에는 못나온다'...수능 출제위원 등 700여명
비공개 합숙 돌입

입소 후 외부 통신·외출 차단, 수능 당일 퇴소, 보안 강화

Design Issues in Offline Recommender Evaluation

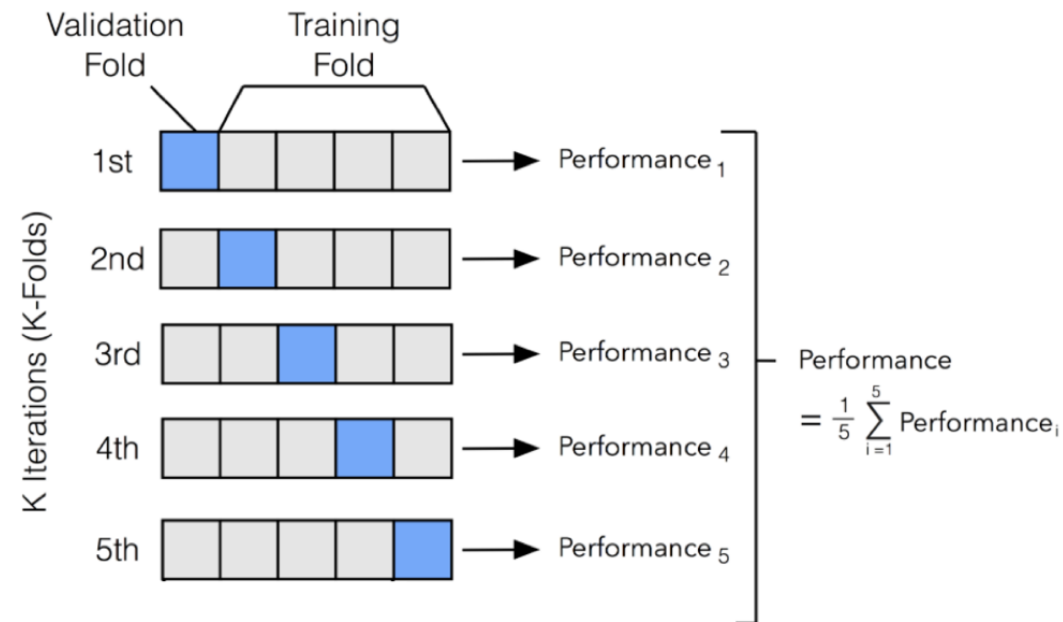
- **Validation Set**

- Test data should never be accessed until the performance of the recommendation model is being evaluated
- Therefore, the **validation set** should be used when selecting the trained model for final evaluation
- The validation set is typically separate from the training set, and the closer it is to the test set, the more accurate the model selection becomes

Design Issues in Offline Recommender Evaluation

- **K-Fold Cross-Validation**

- A validation technique in which the **data is divided into K subsets**, with one subset used as validation data and the remaining as training data in each iteration iteratively to assess the model's performance and estimate the average performance



Design Issues in Offline Recommender Evaluation

- **K-Fold Cross-Validation**

- A validation technique in which the **data is divided into K subsets**, with one subset used as validation data and the remaining as training data in each iteration iteratively to assess the model's performance and estimate the average performance
- Data scarcity can make it challenging to assess the model's generalization performance
- However, K-fold cross-validation allows for more effective utilization of available data to evaluate the model's performance